RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Thursday, 13 August 2026, 11:23 AM
State	Finished
Completed on	Thursday, 13 August 2026, 11:40 AM
Time taken	17 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int arr[], int left, int right, int key) {
    if (left > right)
        return -1;

    int mid = left + (right - left) / 2;

    if (arr[mid] == key)
        return mid;

    if (arr[mid] > key)
        return binarySearch(arr, left, mid - 1, key);

    return binarySearch(arr, mid + 1, right, key);
}

int findPair(int arr[], int left, int right, int x,
             int *a, int *b) {


    if (left >= right)
        return 0;

    int mid = left + (right - left) / 2;


    if (findPair(arr, left, mid, x, a, b))
        return 1;


    if (findPair(arr, mid + 1, right, x, a, b))
        return 1;

    for (int i = left; i <= mid; i++) {
        int complement = x - arr[i];

        int pos = binarySearch(arr, mid + 1, right, complement);

        if (pos != -1) {
            *a = arr[i];
            *b = arr[pos];
            return 1;
        }
    }

    return 0;
}

int main() {
    int n, x;

```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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